

Company news component — project report

A standalone backend that keeps a news feed for every company in a **1,515-company catalogue**. Its own process, its own database, no UI.

Every figure comes from one run and is reproducible with `npm run metrics`.

Results

Companies in catalogue	1,515	Companies returning news	1,005 (66.3%)
Resolved to a real security	1,463 (96.6%)	Attribution certain	99.8% of links
Exchange listings found	1,987	Articles carrying an image	82.0%
Depository receipts found	140	Refusals / failures	0 / 0

Recurring cost: \$0. No paid API, and no language model is called anywhere.

What each row means

- **Resolved to a real security** — for 96.6% of tickers we worked out exactly which real, tradeable share it refers to.
- **Attribution certain** — 99.8% of stored articles came from a provider that *knew the ticker*. Only 0.2% rest on matching a company name.
- **Exchange listings** — those 1,463 companies trade in 1,987 places in total; many trade on more than one exchange.
- **Depository receipts** — 140 of those listings are a foreign share wrapped for trading elsewhere: 139 American (ADR) and 1 global (GDR).
- **Refusals / failures** — no provider refused us, and nothing crashed.

The idea that makes it work

The problem: free news APIs only cover companies listed in America. A third of this list isn't American.

The trick: most foreign companies *also* trade in America under a different code. Toyota is 7203 in Tokyo, but also TM in New York. That American version is a **depository receipt** — the same company, wrapped so Americans can buy it.

So before fetching any news, we find every place each company trades. If any of them is American, a free US news API can cover it.

Segment	Companies	What it means
US exchange listing	1,252	Trades on NYSE or Nasdaq — well covered
US OTC line only	182	An American presence, but on the quieter over-the-counter market, where coverage is thinner
Reachable with a US symbol	1,434 (94.6%)	The two above: we have <i>some</i> American code to ask about
No US line at all	29	Trades only abroad — must fall back to name search
Unresolved	52	Could not identify the company (mostly delisted)

94.6% is the whole argument. Because the listing work was done first, almost everything became reachable with free tools. Without it, we would have had to buy a global news service.

A ticker is not an identifier

A ticker is just a short code **on one particular exchange**. The same code means different companies on different exchanges:

- **BBY in London** = Balfour Beatty, a construction firm
- **BBY in New York** = Best Buy, an electronics retailer

Ask for "news for BBY" and you get **Best Buy's news filed under Balfour Beatty**. The feed looks completely normal — full of real articles — and is about the wrong company entirely. That is what makes it dangerous: nothing looks broken.

We found seven of these:

BBY Balfour Beatty → Best Buy	ADM Admiral → Archer-Daniels	NOV Novo Nordisk → NOV Inc	ENR Siemens Energy → Energizer
FTK flatexDEGIRO → Flotek	CWK Cranswick → Cushman	MOVE Medacta → Corvex	

Filtering by exchange does not fix it. The obvious fix is to ask for "BBY, *in London*". That fails too, because Archer-Daniels-Midland (ADM in New York) **also trades in London** — so "ADM in London" still returns the wrong company.

The only reliable check is the name. We ask the provider what company it thinks the ticker is, then compare that to the name in our list. If they disagree, we reject it.

All seven now resolve correctly — and each **gained a US symbol in the process**. While correctly identifying Balfour Beatty we also found its American code, BAFBF , so its news can now come from a free US provider.

Wrong articles are thrown away, not hidden

For companies we can look up **by ticker**, the provider tells us exactly which company a story is about — no guessing. But for companies with no US listing there is no ticker to ask with, so we fall back to **searching by company name** — and a name search cannot tell a company called Jensen from a person called Jensen.

Jensen-Group is a Belgian maker of industrial laundry machines; its name shortens to "Jensen". **Jensen Huang** is the CEO of Nvidia, one of the most written-about people in business. Searching for "Jensen" returned hundreds of Nvidia stories, filed under a small Belgian laundry-equipment maker.

Same pattern with **Kid ASA**, a Norwegian home-textiles retailer. Its name shortens to "Kid"; searching returned a review of a film called *Club Kid*.

What we do about it. Before saving any article found by name search, we check it is genuinely about that company:

1. **Reduce the name to its distinctive words** — Jensen-Group → jensen .
2. **Multi-word names need two words present.** "Munich Reinsurance lifts profit guidance" keeps; "Munich hosts Oktoberfest" is rejected.
3. **Single-word names need the word *and* business vocabulary** — shares, earnings, revenue, results, acquisition, dividend. "Jensen Huang could redefine NVIDIA" is rejected; "Jensen Group reports record revenue" keeps.

Articles that fail are **never stored** — not saved and quietly marked "low quality". **3,041 were thrown away in the last run.** A wrong article never reaches the feed at all.

Which companies have no news, and why

The full list is exported as data — **data/coverage-gaps.csv** , one row per company with a gap_reason , regenerated with `npm run export:gaps` . **510 of 1,515** produced no news. They are not one problem:

Gap reason	Companies	Whose problem is it?
no_news_us_exchange	276	Nobody's. Identified correctly, asked correctly, genuinely quiet that week
no_news_otc_only	154	Thin OTC coverage — a paid provider would help
no_news_no_us_line	29	Trades only abroad; no free US-centric source reaches it
unresolved_london_secondary_line	16	Ours — 0XXX LSE codes are not in our identifier sources
unresolved_frankfurt_line	13	Ours — same, for .F Frankfurt lines
unresolved_absent_from_directory	10	Absent from Finnhub's US directory
duplicate_listing_line	7	Nothing is missing — same company already covered under another ticker
unresolved_name_check_rejected	3	Ours — the safety check fired, twice wrongly
unresolved_malformed_ticker	2	The catalogue's — LVMH.F uses _ where every other row uses .

A correction worth stating plainly. An earlier draft of this report said the 52 unresolved were "45 delisted, which is correct." That was overconfident. What is actually true is narrower: **45 could not be matched to an identifier in the sources we use**. Checking them individually shows most are not delisted at all:

- **16 are London 0XXX lines** — 0FIN Orkla, 0IU8 Safran, 0NQM Vinci, 0R22 Barrick Gold. Large, actively traded companies. The 0XXX code is an LSE convention our identifier sources do not index.
- **13 are Frankfurt .F lines** — EADS.F Airbus, MAKS.F Marks & Spencer, CMXH.F CSL. Same story.
- **7 are duplicates** — EADS.F Airbus is already covered as 0KVV ; RYAA.Y Ryanair as 0RYA . Nothing is missing for these at all.

- **10 are plain US tickers absent from Finnhub's 30,995-symbol US directory** — EA, SKX, EQR, JHG. Some are genuine take-privates; others look like directory gaps. We do not claim to know which without checking each.

Two of the three name-check rejections are our bug, not a bad match. WAB was rejected because the catalogue says "Westinghouse Air Brake Technologies" and the directory says "WABTEC CORP" — the same company under its trading name. MUV2 was rejected because the directory carries Munich Re's German name, "MUENCHENER RUECKVER AG-REG". The matcher compares surface forms and has no notion of an abbreviation or a translation. The third rejection, P → "EVERPURE INC-A" against a catalogue row for Pure Storage, is **correct** — those really are different companies, and it is the same protection that caught the seven ticker collisions.

The catalogue also double-counts. 1,515 rows are not 1,515 distinct businesses: Novartis appears three times (NOVN, NOVNEE, 0QLR), SAP three times (SAP, 0NW4, SAPG.F), Alphabet twice (G00GL, 0HD6). Coverage percentages are therefore computed per row, and the true per-business figure is somewhat better than the headline suggests.

What is actually fixable

Fix	Companies recovered	Effort
Treat _ as . when parsing tickers	2	Trivial
Alias table for trade names and non-English names	2	Small, manual
Collapse duplicate listing rows onto one company	7	Small
Map 0XXX / .F codes via an LSE/Deutsche Börse identifier source	29	Medium
Paid provider for OTC and non-US lines	up to 183	Cost, not code

The first three are ~11 companies for an afternoon's work. The last row is the only one that requires spending money, and it is the largest by far.

Every link now points at the publisher

Two separate faults made stored links fail when clicked. Both are fixed.

1. Google News redirects (~1,400 articles). These arrived as news.google.com links rather than publisher addresses, and cannot be repaired: the token decodes to an opaque Google identifier rather than the article's address, the page returns nothing to a non-browser client, and browsers frequently refuse the redirect. Such a story can never show a picture either. We stopped storing them.

2. Finnhub redirect wrappers (4,975 articles — 98% of the feed). Finnhub does not return the article's URL. It returns a link into its own domain, finnhub.io/api/news?id=..., which 302s onward to the publisher. Storing that was a mistake on three counts:

- every reader paid an extra hop, and the link dies entirely if Finnhub is down — a news archive should not depend on the liveness of the API it came from;
- the wrapper **hid the destination**, so a dead publisher could not be filtered: the URL looked healthy right up until the click;
- **dedupe degraded**, because every wrapper carries a distinct id, so one wire story syndicated to three outlets was three separate articles.

Ingest now resolves the wrapper to the publisher's own address before storing, and a backfill rewrote all 4,975 existing rows. **0 wrappers remain**, across **76 distinct publishers**.

What the wrapper was hiding: chartmill.com. With destinations visible, one host turned out to be dead — it refuses the connection outright, returning no HTTP status at all in ~0.35s, with a browser user-agent, which a reader sees as a 504. It accounted for 189 articles and 188 broken images. It is now filtered, via a configurable DEAD_ARTICLE_HOSTS list.

That list is deliberately tiny and evidence-based. It is **not** a list of sites that block automated clients: Benzinga and Seeking Alpha both refuse our requests while serving browsers perfectly well, and dropping them would discard 2,298 working articles. A host qualifies only when it fails for a real browser too.

	Before	After
Links via a redirect wrapper	4,975 (98%)	0
Links on dead hosts	189	0
Broken Google links	~1,400	0
Articles carrying an image	65%	82.0%
Companies returning news	1,163	1,005

The lost coverage is real, and it is the clearest argument for a paid provider: one returns genuine publisher links and images for exactly these companies.

The two timing numbers

They measure different things.

A 26-minute run — fetching news for every company, day to day. Finnhub's free plan allows 60 requests per minute, and we ask about one company per request. $1,434 \text{ companies} \div 55 \text{ per minute} \approx 26 \text{ minutes}$. **The code is not slow — it is deliberately waiting** to stay inside the free limit. A paid tier would cut this to a few minutes and change nothing else.

A ~9-minute first start — this happens **once**. On first boot we look up all 1,515 tickers to find their exchanges, via OpenFIGI. Without a key that allows 25 lookups a minute; with a **free** key, 250 a minute and larger batches — hence 9 minutes rather than 38. After the first time it never happens again.

Design decisions worth knowing

- **Provider-agnostic.** Sources sit behind one interface, ordered by config. Adding one is a config change, not a rewrite.
- **Idempotent.** Articles are keyed on a cleaned-up URL, so a daily re-run stores nothing new. Verified: zero duplicate URLs.
- **"No news" is not "provider refused."** They look identical if you only count articles. The first is normal; the second means coverage is dying. They are recorded separately, along with every provider tried per company — because a working fallback can hide a broken primary.
- **No language model.** Article text is written by strangers; letting a model read it and decide what gets stored would hand outsiders influence over the pipeline. The rules used instead are unit-tested against real failures.

Do we need to pay?

Not for coverage — 94.6% is already reachable free and ticker-native.

Yes, if two things matter. First, the ~170 companies whose only links were unopenable or whose OTC coverage is thin. Second, **provenance**: most company news comes from a handful of aggregators rather than primary wires.

Marketaux was evaluated on its free tier. It returns genuine publisher links and images and has real depth for major international names (4,960 articles for Legal & General), but it does not cover Chinese A-shares. Its adapter is built and activates on an API key, so the decision is reversible at any time.

Known limits

- **52 companies (3.4%) unresolved** — see the table above; most are foreign secondary listings our identifier sources do not index, not delistings.
- **Name matching is the weakest component.** Some pairs are undecidable from names alone: "Adobe Systems" vs "Adobe" is indistinguishable from "Prudential" vs "Prudential Financial" — the first is one company, the second is two. Those route to a flagged, reduced-confidence path rather than a guess.
- **Some publishers block us.** Bloomberg returns an error to automated requests, so those stories carry no image. We do not disguise the service as a browser to get around it.
- **One ingest worker**; parallel workers need shared rate limiting.

Full detail: [README.md](#) · [COMPARISON.md](#) · [API.md](#) · [OPERATIONS.md](#) · [data/listings-mapping.csv](#)